

AdVantage AI — Marketing Campaign Management System

5-Part Team Task Breakdown & Assignment Worksheet

This official assignment worksheet breaks down the full-stack **AdVantage AI** project into 5 modular, balanced parts. Each section assigns dedicated file ownership, technical responsibilities, and key deliverables for smooth team collaboration. **Assigned Member / Role fields have been left blank for custom assignment.**

PART 1: BACKEND API & DATABASE INFRASTRUCTURE

Assigned Teammate:	
File Ownership:	backend/server.js, backend/models/Campaign.js, backend/models/Post.js, backend/models/Profile.js, backend/models/Notification.js, backend/models/History.js

Key Deliverables & Responsibilities:

- Database Schema Design:** Define Mongoose schemas for Campaign, Post, Profile, Notification, and History with strict validations.
- REST API Endpoints:** Build endpoints for /api/campaigns (CRUD), /api/posts, /api/profile, /api/history, and /api/generate.
- Middleware & Seed Fallbacks:** Configure CORS, body-parser middleware, error logging wrappers, and offline database fallback objects.

PART 2: UI FOUNDATION, THEME & MAIN DASHBOARD

Assigned Teammate:	
File Ownership:	src/App.jsx, src/index.css, src/components/Sidebar.jsx, src/components/Topbar.jsx, src/components/Dashboard.jsx

Key Deliverables & Responsibilities:

- Design System & Theme Engine:** Maintain glassmorphic layout, modern color palette, and Dark/Light mode theme state.
- Core Navigation & Topbar:** Build Sidebar navigation tabs and Topbar notification drawer + profile quick menu.
- Main Dashboard Control Room:** Render key metric stat cards (Active Campaigns, Est. ROI, Posts, Channels) and recent activity feed.

PART 3: CAMPAIGN MANAGER, ANALYTICS & AUDIT HISTORY

Assigned Teammate:	
File Ownership:	src/components/Campaigns.jsx, src/components/Analytics.jsx, src/components/HistoryLog.jsx, src/api.js

Key Deliverables & Responsibilities:

- **Campaign Management:** Build creation modal & form (Objective, Budget, Schedule, Channels), edit, delete, and filter controls.
- **Chart.js Analytics:** Integrate Line charts (weekly trends), Polar Area (channel distribution), and Bar/Doughnut (budget vs ROI).
- **Audit Log History:** Render chronological timeline of system modifications with real-time category search filters.

PART 4: AI COPY GENERATION ENGINE & EMAIL CREATOR

Assigned Teammate:	
File Ownership:	src/components/ContentPreview.jsx, src/components/EmailCreator.jsx

Key Deliverables & Responsibilities:

- **AI Post Creator:** Build social media drafter (Instagram, LinkedIn, Twitter, Facebook) with automatic post captioning & hashtags.
- **Pollinations AI API Integration:** Connect external LLM API with intelligent offline fallback rules so zero downtime occurs.
- **Interactive Email Creator:** Build dynamic HTML email template generator (Welcome, Promo, Launch) with live preview sandbox.

PART 5: USER PROFILE SETTINGS, QA, PERFORMANCE & DEPLOYMENT

Assigned Teammate:	
File Ownership:	src/components/Settings.jsx, SRS_Document.md, README.md, .env

Key Deliverables & Responsibilities:

- **User Profile Settings:** Build profile form with Base64 avatar uploader and live sync to Topbar header.
- **QA & Performance Testing:** Enforce form validations, fast Vite compilation (<1.5s), and mobile/tablet responsive layouts.
- **Documentation & Deployment:** Setup environment variables (.env), npm scripts, deployment guides, and team review.

■ Git Workflow & Collaboration Guidelines

- **Branching Strategy:** Each part lead works on a feature branch (e.g., feature/part-1-backend, feature/part-2-ui-dashboard).
- **API Agreement:** All API routes prefix /api/ and return standard JSON structure: { success: true, data: {...} }.
- **Testing:** Test backend routes with Postman/ThunderClient and verify Vite frontend rendering before merging to main.